

Leg-haemoglobin is found in root nodules of leguminous plants such as alfalfa and soybean. It is crucial for supplying sufficient oxygen to root nodule bacteria for nitrogen fixation to occur.

28. Which among the following animal is called as—
'Ploughers of the soil'?

- (a) Earthworm (b) Snake
(c) Frog (d) None of these

Uttarakhand P.C.S. (Pre) 2021

Ans. (a)

The earthworms are called the 'Ploughers of the soil'. The earthworms act like ploughs by naturally ploughing our soil and aerating it as well as letting in the necessary supply of air and water, helping plants to grow. They even help by breaking down organic matter, like leaves and grass by converting them into things that plants can use.

Human Anatomy and Physiology

I. Skeletal and Muscle System

Notes

- **Anatomy** is the branch of biology concerned with the shape, structure and relationship of parts of the organism.
- **Skeletal and Muscle system** serves as a framework for the body of all vertebrates including human.
- The skeletal system composed of bones, cartilages, ligaments and other tissues that perform essential functions of the body.
- **Bone tissue** or **Osseous tissue** is the major structural and supportive connective tissue of the body. It forms the rigid part of the bones that make up the skeletal system. Bone tissue is different from the bones themselves - bones are organs made up of bone tissue as well as marrow, blood vessels, epithelium and nerves. It helps in movement and protect the internal organs of the animals.
- **Cartilages**, a resilient and semirigid connective tissue found on the skeleton in areas of flexibility and smooth surface support movement.
- **Ligaments** are the fibrous and dense connective tissues, that connect bones to other bones.
- **Tendon** (or sinew) is a tough band of fibrous connective tissue that connects muscle to bone, or muscle to muscle and is designed to withstand tension.
- The major functions of the skeletal system are body support, facilitation of movement, protection of the

internal organs, storage of mineral and fat, and blood cell formation.

- Bones serve as a site for fat storage and blood cell production. The unique semi-solid tissue that fills the interior of most bones is referred to as **bone marrow**.
 - There are two types of bone marrow : (i) **Yellow bone marrow** which contains **adipose tissue** and triglycerides stored in the adipocytes of this tissue can be released to serve as a source of energy for other tissues of the body and (ii) **Red bone marrow**, where the production of blood cells (named hematopoiesis, hemato = "blood", poiesis = "to make") takes place. Red blood cells, white blood cells and platelets are all produced in the red bone marrow.
 - The **skeleton** has been divided into two parts :
 1. **Exoskeleton** - a hard outer layer that covers, supports and protects the body of an invertebrate animal such as an insect or crustaceans which is made up of **chitin**.
 - The exoskeleton of vertebrates includes scales in fishes and reptilians, feathers in birds and hair, nails, horns, hoofs in mammals. These are made of α -keratin protein.
 2. **Endoskeleton** - a hard frame that supports the body of a vertebrate animal on the inside, which is made of **bones** and **cartilages**.
 - (i) **Bone** : Bone is the substance that forms the skeleton of the body. It is composed chiefly of **calcium phosphate** and **calcium carbonate**. It also serves as a storage area for calcium, playing a large role in calcium balance in the body. Collagen is the main structural protein which is found in the organic matrix of bone (and various connective tissues in the body).
 - (ii) **Cartilage** : Cartilage is a resilient and smooth elastic (but semirigid) tissue, a rubber-like padding that covers and protects the ends of long bones at the joints, and is a structural component of the rib cage, the ear, the nose, the bronchial tubes, the intervertebral discs, and many other body components. It is much stiffer and much less flexible than muscle. The matrix of cartilage is made up of **glycosaminoglycans**, **proteoglycans**, **collagen** fibres and, sometimes, elastin. Cartilage does not contain blood vessels (avasular) or nerves (aneural). Nutrition is supplied to the chondrocytes (the only cells found in healthy cartilage) by diffusion.
- Endoskeleton of Human :**
- The endoskeleton of human is an internal skeleton composed of hard, mineralized tissues that also enables movement by attachment of muscles.

- In adult human endoskeleton, 206 bones are found. It is comprised of the **axial** and **appendicular** skeleton. Short details are given as below-

(i) Axial Skeleton :

- Axial Skeleton of the human comprises of-
- (a) Skull :** There are basically 22 bones (8 - cranial, 14 - facial) in human skull. Including the hyoid (in the neck) and the bones of the middle ears, the head contains 29 bones. In general classification, these 29 bones are all considered as part of skull. These are as follows :
 - (i) Cranium – 8
 - (ii) Face – 14
 - (iii) Ears – 6 (three in each ear)
 - (a) Malleus – hammer (2)
 - (b) Incus – anvil (2)
 - (c) Stapes – stirrup (2)
 - (iv) Hyoid - 1

Stapes is the smallest bone in human

(b) Vertebral Column : The vertebral column usually consists of 26 vertebrae as follows :

- (i) Cervical Vertebrae – 7 (C1 to C7; its first vertebra-C1 is called as atlas and second-C2 as axis)
 - (ii) Thoracic Vertebrae – 12
 - (iii) Lumbar Vertebrae – 5
 - (iv) Sacral Vertebra – 1 (5 fused sacral vertebrae)
 - (v) Coccyx Vertebra – 1 (4 fused sacral vertebrae)
- (c) Ribs :** 12 paired ribs (total - 24), each is symmetrically paired on a right and left side.
- Our **thoracic cage** or **rib cage** is composed of 12 pairs of ribs, sternum, cartilages and thoracic vertebrae. Of the 12 pairs of ribs the first 7 pairs are attached to vertebrae posterior and sternum anteriorly (with costal cartilage). These rib pairs are called **true ribs**.
 - No. 8-12 rib pairs are attached to vertebrae posteriorly but anteriorly they are not attached to the sternum directly. These rib pairs are called **false ribs**.
 - The 8-10th rib pairs connect to the sternum indirectly via the costal cartilages of the ribs above them.
 - The 11 & 12th rib pairs do not connect to the sternum at all. So these two rib pairs are called **floating ribs**.
- (d) Sternum :** The sternum or breastbone (1) is a long flat bone located in the central part of the chest. It serves as the keystone of the rib cage and stabilizes the thoracic skeleton.

Total number of Bones in Axial skeleton - 80

(ii) Appendicular Skeleton :

- The appendicular skeleton of a human includes the bones of the shoulder girdle, upper limbs, pelvic girdle and the lower limbs.
- (a) Pectoral Girdle (Shoulder Girdle) :** Total 4 bones
 - (i) Scapula (Shoulder blade) – 2
 - (ii) Clavicle (Collarbone) – 2
- (b) Pelvic Girdle :** 2 bones
 - (i) Os innominatum (Hip bone) – 2
- (c) Forelimb :** Total 60 bones
 - (i) Upper arm (Humerus) – 2 (one in each forelimb)
 - (ii) Forearm (Radioulnar) – 4 (two in each forelimb)
 - (iii) Wrist (Carpal) – 16 (8 in each forelimb)
 - (iv) Palm (Metacarpal) – 10 (5 in each forelimb)
 - (v) Fingers (Phalanges) – 28 (14 in each forelimb)
- (d) Hindlimb :** Total 60 bones
 - (i) Thigh (Femur) – 2 (1 in each hindlimb)
 - (ii) Shank (Tibiofibula) – 4 (2 in each hindlimb)
 - (iii) Kneecap (Patella) – 2 (1 in each hindlimb)
 - (iv) Ankle (Tarsal) – 14 (7 in each hindlimb)
 - (v) Foot (Metatarsal) – 10 (5 in each hindlimb)
 - (v) Fingers (Phalanges) – 28 (14 in each hindlimb)

Total number of Bones in Appendicular skeleton - 126

Femur (Thigh bone) is the longest bone of human skeleton

Arthritis :

- The word arthritis is used to describe pain, swelling and stiffness in a joint or joints.
- **Osteoarthritis** is the most common type of arthritis. When the cartilage - the slick, cushioning surface on the ends of the bones - wears away, bones rub against bone, causing pain, swelling and stiffness. It can affect both the larger and smaller joints of the body, including the hand, wrists, feet, back, hip and knee.
- **Rheumatoid arthritis (RA)** is a disorder in which the body's own immune system start to attack body tissue. It typically results in warm, swollen and painful joints. In RA, most damage occurs to the joint lining and cartilage which eventually results in erosion of two opposing bones. RA often affects joints in the knees, wrists, elbows and fingers, is symmetrical (appears on both sides of the body), and can lead to severe deformity in a few years, if not treated.
- **Metabolic arthritis (Gout)** is that when **uric acid** is formed as the body breaks down purines, a substance found in animal cells and in many foods. The crystals of

uric acid are deposited in the joint, resulting in sudden spikes of extreme joint pain or a gout attack.

- **Arthroplasty** is an **orthopaedic surgical procedure** where the articular surface of a musculoskeletal joint is replaced, remodeled or realigned by **osteotomy** or some other procedure. It is an elective procedure that is done to relieve pain and restore function to the joint after damage by arthritis or some other type of trauma.

Water Percentage in the Human Body :

- The percentage of water varies generally between about 55-70% according to the age and gender of human.
- Average Water % in adult men – about 60-65%
- Born infants – about 75-78%
- Infants by one year of age – about 65-70%
- In adult women, fats make up more of the body than men, so they have about 55-60% of water in their bodies.
- According to H.H. Mitchell, Journal of Biological Chemistry :
 - The brain and heart composed of 73% water,
 - Lungs are about 83% of water,
 - Skin contains 64% water,
 - Muscles and Kidneys contain 79% water, and
 - Bones contain 31% of water.
- An adult male needs about 3 Litres water per day while an adult female needs about 2.2 litres water per day.
- Water serves a number of essential functions to keep us all going :
 - A vital nutrient to the life of every cell, acts first as a building material.
 - It regulates our internal body temperature by sweating and respiration.
 - The carbohydrates and proteins that our body uses as food are metabolized and transported by water in the bloodstream.
 - It assists in flushing waste mainly through urination.
 - It acts as a shock absorber for brain, spinal cord and fetus,
 - forms saliva,
 - lubricates joints.

Chemical Composition of Human Body :

- Almost 99% of the mass of human body is made of six elements : **oxygen, carbon, hydrogen, nitrogen, calcium and phosphorus**.
- Only about 0.85% is composed of another 5 elements : potassium, sulphur, sodium, chlorine and magnesium.
- All 11 elements mentioned above are necessary for life.
- The approximate percentage of different elements in human body is shown in following table :

Element	Symbol	% in body
Oxygen	O	65.0
Carbon	C	18.5
Hydrogen	H	9.5
Nitrogen	N	3.2
Calcium	Ca	1.5
Phosphorus	P	1.0
Potassium	K	0.4
Sulphur	S	0.3
Sodium	Na	0.2
Chlorine	Cl	0.2
Magnesium	Mg	0.2

- Trace elements including boron, chromium, cobalt, copper, fluorine, iodine, iron, manganese, molybdenum, selenium, silicon, tin, vanadium and zinc are less than 0.1% of the human body.

Calcium phosphate compound is found in maximum amount in human bone

Human Tooth :

- The human teeth function to mechanically breakdown items of food by cutting and crushing them in preparation for swallowing and digesting.
- Humans have four types of teeth :
 - Incisors – cut the food
 - Canines – tear the food
 - Premolars and Molars – crush or grind the food
- The roots of the teeth are embedded in the **maxilla** (upper jaw) or the **mandible** (lower jaw) and are covered by gums.
- Human teeth are **diphyodont** meaning that they develop two sets of teeth. The first set (called baby ‘milk’, primary or deciduous set), normally starts to appear to about six months of age, although some babies are born with one or more visible teeth, known as **natal teeth**.
- Normal tooth eruption at about six months is known as teething and can be painful.
- Humans usually have 20 primary teeth and 32 permanent (adult) teeth.
- Among permanent teeth, 16 are found in maxilla and 16 in the mandible.
- Third molars are commonly called ‘**wisdom teeth**’ and may never erupt into the mouth or form at all.
- If any additional teeth form (eg. fourth and fifth molars), which are rare, they are referred to as supernumerary teeth (**hyperdontia**).

- Tooth **enamel** is the hard mineralized surface of the teeth. It is the hardest substance of the body and it is whitish and shining substance.
- Essential nutrients for a healthy tooth include calcium, phosphorus, fluorine and vitamin A, D, C.
- Calcium and phosphorus are needed to properly form the **hydroxyapatite crystals** (which is enamel's primary mineral) and their levels in blood are maintained by Vitamin D.
- Demineralization begins with the bacteria in the plaque on teeth. They feed on sugar and other carbohydrates in mouth and produce acids. The acids dissolve crystals in tooth enamel. The loss of enamel is balanced by remineralization with the help of fluorine, calcium, phosphate, etc. Too much loss of minerals without enough replacement leads to tooth decay.

- **Sartorius muscle** is the longest muscle in the human body which runs diagonally down the thigh.
- **Stapedius** is the smallest muscle which is less than 2 millimeter long located in the middle ear.
- **Strongest muscle** in the human body is **masseter**, muscle known as jaw muscle that plays a major role in the chewing of the solid foods.
- Record of lifting 975 pound weight by jaw muscle for 2 seconds is registered in Guinness World Records.

Question Bank

- Weight of our body is mostly made of :**
 - From bones
 - Parts of body
 - Parts of skin
 - From water

Uttarakhand U.D.A/L.D.A. (Pre) 2003

Ans. (d)

The amount of water in the human body generally ranges from 55-70%. The average adult human body have about 55-65% water. The percentage of water in infants is much higher typically around 75-78%, dropping to 65-70% by one year of age. Water is the best solvent in nature.

- Human body contains :**
 - about 70% water
 - 20%-30% water
 - 10%-20% water
 - 30%-40% water
 - None of the above/More than one of the above

64th B.P. S.C. (Pre) 2018

Ans. (a)

See the explanation of above question.

- The percentage of water in an adult human body is around–**
 - 10%
 - 30%

(c) 65%

(d) 75%

U.P.P.C.S. (Mains) 2007

Ans. (c)

See the explanation of above question.

- The most abundant element found in the human body is –**
 - Iron
 - Sodium
 - Oxygen
 - Iodine

U.P.P.C.S. (Mains) 2014

Ans. (c)

Nearly 99% of the mass of human body consists of just six chemical elements– oxygen, carbon, hydrogen, nitrogen, calcium and phosphorus. Oxygen is the most abundant element in the human body. It is mainly found in the form of water. Water, makes up generally about 55-70% of the human body and participates in countless metabolic reactions.

- The maximum amount of element found in human body is–**
 - Iron
 - Sodium
 - Phosphorus
 - Calcium

U.P. U.D.A./L.D.A. (Pre) 2013

Ans. (d)

Almost 99% of the mass of the human body is made up of six elements, these are approximately : Oxygen (65%), Carbon (18.5%), Hydrogen (9.5%), Nitrogen (3.2%), Calcium (1.5%) and Phosphorus (1.0%). So according to question, option (d) is correct.

- Which one of the following salts is found in maximum amount in the human bones?**

- Magnesium Chloride
- Calcium Carbonate
- Calcium Phosphate
- Sodium Chloride

U.P.P.C.S. (Mains) 2015

Ans. (c)

Calcium phosphate is the principal form of calcium found in bovine milk and blood. About 70% of bone consists of hydroxyapatite, a calcium phosphate mineral (known as bone mineral). Up to 50% by volume and about 70% by weight of human bones is a modified form of hydroxyapatite.

- In human body percentage of which element is highest?**
 - Hydrogen
 - Nitrogen
 - Carbon
 - Oxygen

M.P.P.C.S. (Pre) 2019

Ans. (d)

See the explanation of above question.

- The number of bones in an adult human body is –**
 - 204
 - 206

(c) 208

(d) 210

Uttarakhand P.C.S. (Pre) 2010

U.P. Lower Sub. (Pre) 2004

Ans. (b)

The number of bones in an adult human body is 206. The body of a child has about 300 bones at birth, which decreases to 206 bones by adulthood after some bones have fused together. Together these bones form the axial skeleton and the appendicular skeleton.

9. How many bones are there in human skeleton?

(a) 280

(b) 200

(c) 220

(d) 206

R.A.S./R.T.S. (Pre) 1997

Ans. (d)

See the explanation of above question.

10. At the age of 20 years, the number of bones in human body is approximately-

(a) 100

(b) 200

(c) 300

(d) None of these

41st B.P.S.C. (Pre) 1996

Ans. (d)

See the explanation of above question.

11. Which of the following is the smallest bone in the human body ?

(a) Vomer

(b) Stapes

(c) Malleus

(d) Incus

U.P. U.D.A./L.D.A. (Spl) (Pre) 2010

U.P. Lower Sub. (Pre) 2008

Ans. (b)

The stapes is the lightest stirrup-shaped bone and the smallest bone in the human body found in the middle of humans ears. While femur is the largest bone in the human body. It is located in the upper leg which connects the knee at one end and fits into the hip socket at the other.

12. The smallest bone in our body is found in our –

(a) Ear

(b) Nose

(c) Eye

(d) Toe

U.P. Lower Sub. (Spl) (Pre) 2004

Ans. (a)

See the explanation of above question.

13. The number of bones in human skull is –

(a) 28

(b) 30

(c) 32

(d) 40

39th B.P.S.C. (Pre) 1994

Ans. (a)

The human skull (excluding hyoid bone of the neck) consists of 28 bones. It consists of two parts, neurocranium and the facial skeleton (also called the viscerocranium). Neurocranium has 8 bones and there are 14 in the viscerocranium (facial bones) and remaining 6 bones occurs in the middle ear, which is helpful in listening.

14. The number of cervical vertebrae in a human body is:

(a) 5

(b) 7

(c) 12

(d) 33

M.P. P.C.S. (Pre) 2024

Ans. (b)

The human cervical spine comprises 7 vertebrae (C1 to C7) and is divided into 2 major segments. The 2 most cephalad vertebrae, the atlas (C1) and the axis (C2), form the craniocervical junction (CCJ) together with the occiput. The 5 cervical vertebrae caudad, C3 to C7, comprise the subaxial spine and are referred to by number. The cervical spine supports the weight of the cranium and head and neck motion.

15. The type of joint (pivot) between Atlas and Axis is known as :

(a) Saddle joint

(b) Fibrous joint

(c) Cartilaginous joint

(d) Synovial joint

M.P. P.C.S. (Pre) 2022

Ans. (d)

The Atlantoaxial joint is a type of synovial joint that is classified as a biaxial, pivot joint. This joint resides in the upper part of the neck between the first and second cervical vertebrae, also known as the atlas and axis, respectively. It allows for turning of the head from side to side. The pivot joint classification means that it allows for rotation only and is formed by a central pivoting, bony structure that rotates around craniovertebral ligaments.

16. The joint where the neck joins the head is :

(a) Pivotal joint

(b) Saddle joint

(c) Hinge joint

(d) Ball and socket joint

U.P.P.C.S. (Pre) 2024

Ans. (a)

The joint where the neck joins the head is a pivot (pivotal) joint, also known as a rotary joint, allowing for head rotation and some flexion and extension. A pivot joint is a type of synovial joint where a cylindrical bone rotates within a ring-shaped bone, allowing for rotational movement along a single axis.

17. How many pairs of ribs are there in a human body?

(a) 12

(b) 10

(c) 14

(d) 11

M.P.P.C.S. (Pre) 1995

Ans. (a)

The human rib cage is made up of 12 paired (total 24) rib bones. Each are symmetrically paired in right and left side. The ribs are flat, thin bones that together with the sternum make up the rib cage. The ribs provide protection for vital organs in the upper body, including the heart and lungs.

18. In human body, leg bone is?

- (a) Hollow (b) Porous
(c) Solid (d) Rivet

U.P.P.C.S. (Pre) 1994

B.P.S.C. (Pre) 1994

Ans. (a)

Each human long bone (including leg bone) is composed of a shaft (diaphysis) with two flared ends (epiphysis). The diaphysis resembles a hollow cylinder. It is made of hard compact bone that is resistant to bending. The inner cavity of the diaphysis contains yellow bone marrow that stores fat. The epiphysis is a thin shell of compact bone filled with a lattice or sponge-like structure that is surrounded by red bone marrow (which makes red blood cells).

19. Osteoclasts are associated with which of the following?

- (a) Bone formation (b) Basal lamina secretion
(c) Bone fracture (d) Muscle regeneration

M.P. P.C.S. (Pre) 2023

Ans. (a) & (c)

An osteoclast is a type of bone cell that breaks down bone tissue. This function is critical in the maintenance, repair, and remodeling of bones of the vertebral skeleton. Osteoclasts are the cells that degrade bone to initiate normal bone remodeling and mediate bone loss in pathologic conditions by increasing their resorptive activity. They are derived from precursors in the myeloid/ monocyte lineage that circulate in the blood after their formation in the bone marrow. These osteoclast precursors (OCPs) are attracted to sites on bone surfaces destined for resorption and fuse with one another to form the multinucleated cells that resorb calcified matrixes under the influence of osteoblastic cells in bone marrow. Hence, both options (a) and (c) are correct.

20. What is arthroplasty in surgery :

- (a) Open heart surgery (b) Kidney transplant
(c) Hip-joint replacement
(d) Blood transfusion

U.P.P.C.S.(Pre) 2001

U.P.U.D.A./L.D.A. (Pre) 2002

Uttarakhand P.C.S. (Pre) 2010

Ans. (c)

Hip arthroplasty or hip-joint replacement surgery involves replacing components of the hip-joint with a synthetic implant, to repair the damaged part that is causing pain.

21. The parts of the skeleton that are not as hard as bones and which can be bent are called as :

- (a) vertebrae (b) carpals
(c) cartilage (d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (c)

Cartilage is a strong, flexible connective tissue that protects our joints and bones. Cartilage is an important structural component of the body. It is a firm tissue but is softer and much more flexible than bone and can be bent. Outer ear and tip of the nose are examples of cartilage. On the other hand, the carpal bones are the eight small bones that make up the wrist (or carpus) that connects the hand to the forearm. These bones are hard and will break if bent. Hence, option (c) is the correct answer.

22. The bones are held together by elastic tissues are called–

- (a) Ligaments (b) Fibrous tissues
(c) Fibrin (d) Muscular tissues

U.P. Lower Sub. (Spl.) (Pre) 2008

Ans. (a)

A ligament is a tissue that connects two bones to form a joint. It is fibrous and dense connective tissue.

23. Plaques formed on teeth are made up of –

- (a) Food particles only
(b) Food particles plus saliva
(c) Food particles plus saliva plus mouth acids
(d) Food particles plus saliva plus mouth acids plus bacteria

U.P.P.C.S. (Mains) 2004

Ans. (d)

Dental plaque is a biofilm or mass of bacteria that grows on surfaces within the mouth. It appears as a white/pale yellow 'slime layer' commonly found in between the teeth and along the cervical margins. Plaques formed on teeth are made up of food particles, saliva, mouth acids and bacteria in the oral cavity. Saliva has a normal pH range of 6.2-7.6 with 6.7 being the average pH.

24. No pain is felt while cutting the nails because -

- (a) Nails are made of the material of dead cells in which there is no blood circulation.
(b) Nails are a useless part of the body.
(c) Nails are made of calcium phosphate.
(d) None of these.

M.P.P.C.S. (Pre) 1994

Ans. (a)

Nails are usually made up of thick and multiple layers of a type of protein known as α -keratin and are composed of dead tissues. They are devoid of any nerves and blood vessels and therefore we do not feel any pain while the nails are being cut.

25. Alpha-keratin is a protein, present in :

- (a) Blood (b) Skin
(c) Lymph (d) Eggs

U.P.C.S. (Mains) 2003

I.A.S. (Pre) 1997

Ans. (b)

Alpha-keratin is a family of fibrous structural proteins. It is the key structural material making up the outer (epidermis) layer of human skin. It is the primary component of hair, nails, hooves, and it provides the necessary strength and toughness for masticatory organs, such as the tongue and the hard palate.

26. Hair is composed of protein called –

- (a) Globulin (b) Mucin
(c) Keratin (d) Casein

U.P.S.C. (R.I.) 2014

Ans. (c)

Hair is composed of the protein keratin, which is also the primary component of finger and toe nails.

27. Which one of the following proteins is present in human hair and nails?

- (a) Oxytocin (b) Keratin
(c) Vasopressin (d) Trypsin

U.P.R.O./A.R.O (Pre) 2014

Ans. (b)

Keratin is an extremely strong protein that is a major component in skin, hair, nails, hooves, horns and teeth.

28. Which of the following are required for the formation of bones and teeth ?

- (a) Sodium and Potassium (b) Iron and Calcium
(c) Sodium and Calcium (d) Calcium and Phosphorus

U.P.C.S. (Mains) 2011

Ans. (d)

Teeth and bones both are hard, white and heavy. Teeth are composed of calcium, phosphorus and other minerals. Bones contain calcium, phosphorus, sodium and other minerals, but mostly consist of the protein collagen.

29. Which one of the following nutritional elements is not essential for the formation and strengthening of bones and teeth?

- (a) Calcium (b) Phosphorus

(c) Fluorine

(d) Iodine

U.P.C.S. (Pre) 2015

Ans. (d)

Calcium, phosphorus, and fluorine are essential elements for the formation and strengthening of bones and teeth. Deficiency of these nutritional elements may cause teeth and bone diseases, while iodine is used to control thyroid gland.

30. The number of teeth that are developed twice in the lifetime of human being is

- (a) 4 (b) 12
(c) 20 (d) 28

39th B.P.S.C. (Pre) 1994

Ans. (c)

In human life span, 20 teeth developed twice. Like a human, most of the mammals are diphyodont – It means our 12 permanent molars are monophyodont and others diphyodont. There are four types of teeth– Incisors, canines, premolars and molars.

31. Which of the following is the strongest part of our body ?

- (a) Bone (b) Enamel
(c) Dentin (d) Cementum

Uttarakhand U.D.A./L.D.A. (Pre) 2003

Ans. (b)

Tooth enamel is the thin outer covering of the tooth and the strongest part of our body. It makes our teeth white and it exists to protect the crowns of healthy teeth. Tooth decay is due to the destruction of tooth enamel. It is made up of hydroxyapatite crystals, which is mainly calcium phosphate.

32. Which one of the following statements is incorrect about muscles in human body?

- (a) Muscles can only push the bone.
(b) Muscles work in pairs.
(c) Muscles become shorter, stiffer and thicker when contracted.
(d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (a)

Muscles allow a person to move, speak, and chew. They control heartbeat, breathing, and digestion. Other seemingly unrelated functions, including temperature regulation and vision, also rely on the muscular system. Muscles have a range of functions from pumping blood and supporting movement to lifting heavy weights or giving birth. Muscles work by either contracting or relaxing to cause movement. Muscles work in pairs and become shorter, stiffer and thicker when contracted. So, statement of option (a) is incorrect.